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Hotel Room Reservation Application (OOP Project)

Abstract

The Hotel Room Reservation System is a Java-based application developed using Object-Oriented Programming (OOP) concepts. The purpose of this project is to simplify hotel room booking, manage room availability, and calculate billing automatically. In traditional systems, hotel management is often done manually, which can lead to errors and inefficiency. This project provides a computerized solution to overcome these problems.

The system uses a Room class to represent individual rooms and a Hotel class to manage multiple rooms using a 2D array. It allows users to book rooms, cancel bookings, check availability, and calculate total cost based on the number of days and room type. An interface is used for billing operations to demonstrate abstraction.

This project helps in understanding real-world implementation of OOP concepts such as encapsulation, abstraction, and modular design.

Introduction

The Hotel Room Reservation System is designed to automate the process of managing hotel room bookings. In small hotels, room booking is often handled manually, which can result in confusion and errors. This system provides an efficient and reliable way to manage room reservations.

The project is developed using Java and follows Object-Oriented Programming principles. Each room is represented as an object with properties like room number, type, price, and availability. The hotel manages all rooms using a 2D array, where rows represent floors and columns represent rooms.

Features of the system include:

- Booking rooms
- Cancelling reservations
- Viewing room availability
- Calculating bills

This system is simple, efficient, and easy to understand.

Objectives

- Manage hotel room records
- Track room availability
- Reduce manual work
- Implement booking and cancellation system
- Calculate total bill
- Apply OOP concepts

Scope

This project is suitable for small hotels and academic purposes. It can be enhanced in future by adding database support, graphical user interface, and online booking features.

System Requirements

Hardware:

- Computer with minimum 4GB RAM

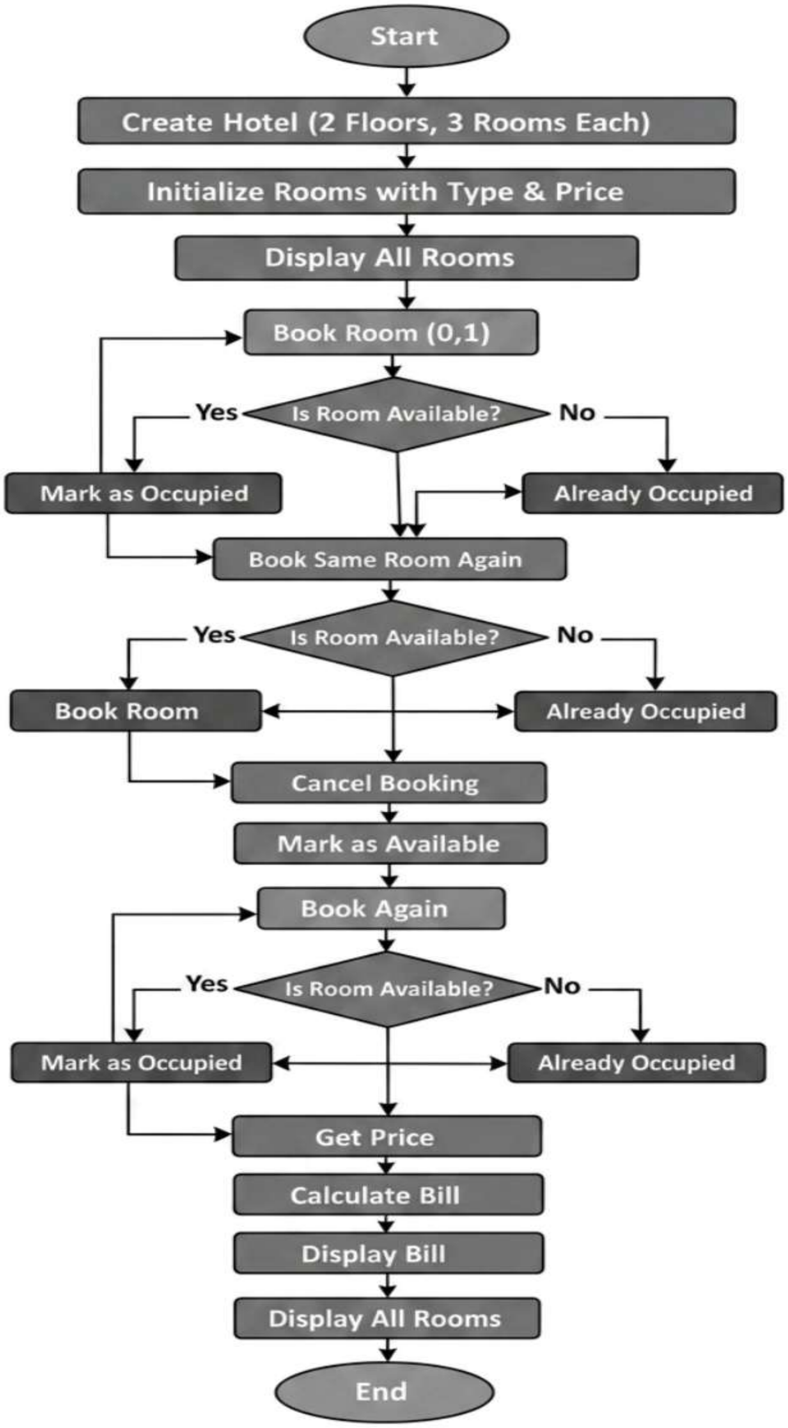
Software:

- Java JDK
- IDE (VS Code / Eclipse / Edit Plus)

Methodology

1. Initialize hotel rooms using a 2D array
2. Assign room type and price
3. User performs operations like booking or cancelling
4. System checks room availability
5. If available → booking successful
6. If not available → show error message
7. Calculate bill based on number of days and room price
8. Display updated room status

Flowchart


Start
Create Hotel (2 Floors, 3 Rooms Each)
Initialize Rooms with Type & Price
Display All Rooms
Book Room (0,1)
Yes
Is Room Available?
No
Mark as Occupied
Already Occupied
Book Same Room Again
Yes
Is Room Available?
No
Book Room
Already Occupied
Cancel Booking
Mark as Available
Book Again
Yes
Is Room Available?
No
Mark as Occupied
Already Occupied
Get Price
Calculate Bill
Display Bill
Display All Rooms
End

Class Diagram

Room Class

- Variables:
    - roomNumber
    - roomType
    - pricePerDay
    - isAvailable
- Method:
    - showRoomDetails()

HOTEL CLASS

- Variables:
    - Room[][] rooms
- Methods:
    - bookRoom()
    - cancelRoom()
    - displayAllRooms()
    - getPrice()
    - calculateBill()

Service Interface

- Method:
    - calculateBill(int days, double rate)

Implementation

The project consists of different classes. The Room class stores the details of each room such as room number, type, price, and availability. The Hotel class manages all rooms using a 2D array and provides functions for booking, cancelling, displaying rooms, and calculating bills. The Service interface is used to define the billing method. The main class controls the program execution and demonstrates all operations.

Sample Input / Output

Room booked successfully

Room already occupied

Booking cancelled successfully

Total Bill for 3 days = 4950.0

Future Enhancements

- Add database connectivity
- Create graphical user interface
- Add online booking system
- Include payment system

- Store customer details

Gantt Chart

Planning – 1 Day

Design – 2 Days

Coding – 3 Days

Testing – 2 Days

Documentation – 1 Day

Conclusion

The Hotel Room Reservation System successfully demonstrates the use of Java and Object-Oriented Programming concepts. It provides an efficient way to manage hotel bookings and billing. The system is simple, structured, and can be expanded with advanced features in the future.

References

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→ Used for understanding core Java concepts and syntax.

2. E. Balagurusamy, Programming with Java, McGraw-Hill.

→ Helped in learning Object-Oriented Programming concepts.

3. Oracle Documentation, Java SE Documentation.

→ Official reference for Java classes, methods, and interfaces.

4. GeeksforGeeks, Java Programming Tutorials.

→ Used for examples and understanding OOP concepts.

Website: https://www.geeksforgeeks.org/

5. W3Schools, Java Tutorial.

→ Used for basic Java syntax and practice.

Website: https://www.w3schools.com/java/

6. College Notes / Classroom Lectures

→ Guidance for project structure and implementation.ye java k a formad ka hai